

User Manual Document

For

Car Manufacturing System

Prepared by

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1. Overview

The Car Manufacturing System (CMS) is a comprehensive, role-based web application that models and manages the complete operational lifecycle of an automobile manufacturing enterprise. It serves as a lightweight, academically-oriented alternative to industrial ERP systems such as SAP, focusing on the core data flows between Product Design, Supply Chain, Production, and Sales.

2. Set up process

Follow these steps to run the project locally.

2.1 Prerequisites:

Make sure you have installed

- Node.js (v16 or higher)
- npm
- PostgreSQL
- Git

2.2 Clone the Repository

```
git clone https://github.com/GV-Kaushik/DBMS-PROJECT.git  
cd DBMS-PROJECT
```

2.3 Frontend Setup

```
cd frontend  
npm install
```

2.4 Backend Setup (Node + Express)

```
cd backend  
npm install
```

2.5 Environment Variables

Create a `.env` file inside the `backend` folder and add:

PORT=5000
DB_HOST=localhost
DB_USER=your_postgres_username
DB_PASSWORD=your_postgres_password
DB_NAME=your_database_name
DB_PORT=5432

2.6 Run Frontend (Run this inside frontend folder)

npm run dev
//Frontend will run on: http://localhost:5173

2.7 Run Backend (Run this inside backend folder)

node server.js
//Backend will run on http://localhost:3000

2.8 PostgreSQL Setup

1. Open PostgreSQL (psql or pgAdmin)
2. Create a database:
CREATE DATABASE cms;
- 3.Import the schema:
Run this in terminal => psql -U postgres -d cms -f cms.sql
in folder DBMS-PROJECT

2.9 API Configuration

Make sure the frontend is calling the backend correctly.

Example: const API_URL = "http://localhost:5000";

2.10 Features

- User Management (CRUD)
 - PostgreSQL Database Integration
 - REST API using Express
 - React Frontend
 - Form Handling & Validation
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2.11 Notes

- Ensure PostgreSQL service is running
 - Check `.env` file for correct DB credentials
 - Backend must run before frontend
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3. Login to CMS:

When you go to the login page, you'll need to enter your details email-ID and password. If the system doesn't find your username, it means you haven't been registered yet. In this case, an admin will handle the registration process for you. The admin will create your account, including your role (admin, supply or design manager, production manager, sales manager) and other necessary details. Once the admin has registered you, you can log in with your credentials. The system will check if your login details are correct. If they are, you'll be able to access the platform. After logging in, your profile will be updated in the system.

By default, an admin user is created in the database with the following credentials:

- Email: admin@gmail.com
- Password: 123

Upon entering admin credentials on Login to Car Manufacturing System will be redirected to the Admin Dashboard page.

Sign out: You can always sign-out from any user session by selecting the sign out option at the left bottom corner of the dashboard to terminate the current session and exit the application.

4. Admin Dashboard:

Upon entering admin credentials on Login to UMS will be redirected to the Admin Dashboard page. The Admin Dashboard serves as the central hub for administrative functions. From here, you can access the following options:

- **Car Models:** Add, modify or delete vehicle model specs, pricing and engine types.
- **Assign Parts:** Assign parts with required quantities to car models.
- **Parts:** Add, modify or delete parts inventory, pricing and their stock.
- **Suppliers:** Add, modify or delete suppliers and their contacts.
- **Supply records:** Track part supply transactions from the suppliers.
- **Production records:** Track production records of the factories by car models.
- **Factories:** Manage factory locations and production capacity.
- **Employees:** Add, modify or delete employees records.

- **Dealers:** Manage dealers and their location information.
 - **Sales:** Track car model sales transactions through dealers.
 - **Manage users:** Add, remove users and assign them roles.
 - **Sign Out:** Terminate the user session and exit the application.
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4.1 Car Models:

This page allows the user (admin) to view all car models, edit their specs like price, name, company and engine , delete existing models or add car models.

4.1.1 Add new Model:

To add a new car model, click the Add Model button in the *Car Models* page and the *Add Car* form will appear. Enter the *Model name*, specify the *company*, *price* and *Engine* of the corresponding car model. Click *Save* after entering the information for adding the new model. Click *Cancel* to return to the previous page without saving.

4.1.2 Edit or Delete Model:

You can edit or delete car models by clicking the respective button present in models row.

After clicking the edit option , *Edit Car* form will appear. You can modify the details of the car model. Click *Save* after entering the information for successfully editing the model. Click *Cancel* to return to the previous page without saving.

To delete a car model , just click the delete option and it will be removed.

Note: When a model is deleted, the records referencing it in Assign Parts also will get deleted.

4.2 Assign Parts:

Select Assign Parts on the Admin dashboard, Assign Parts page will appear. You can see the list of all the parts that are assigned to a car model, and their quantity required.

To assign a new part, click the assign part button on the top-right, fill the form with details of part name, model name, quantity required in that model.

4.3 Parts:

To manage parts, select the Parts option in the dashboard and the Parts page will appear. It contains a list of all the parts along with their category, cost , quantity (in stock).

4.3.1 Adding parts:

Click the add parts option on the top right to add a new part and fill the form with the details of the parts such as quantity, name, category and click *Save* to add it to the database or *Cancel* to exit.

4.3.2 Edit or Delete Parts:

You can edit or remove the parts, by selecting the respective option present on the right side of every part. Edit the parts by modifying details in the Edit Parts form. And click *Save* to make the changes or *Cancel* to return.

Note: You cannot delete parts that have been assigned to any model.

4.4 Suppliers:

To manage suppliers of the parts and their information, select the Suppliers option in the dashboard and the page will appear. It contains a list of all the suppliers with their name, city and contact info.

4.4.1 Adding suppliers:

Click the add suppliers option on the top right to add a new supplier and fill the form with the details of the suppliers such as name, city and contact info , and click *Save* to add it to the database or *Cancel* to exit.

4.4.2 Edit or Delete suppliers:

You can edit or remove the suppliers, by selecting the respective option present on the right side of every supplier. Edit the supplier by modifying details in the Edit Suppliers form. And click *Save* to make the changes or *Cancel* to return.

4.5 Supply Records:

To track the supply records, information of parts and their suppliers, select the Supply Records option in the dashboard and the page will appear. It contains the list of all records with parts, supplier, quantity.

4.5.1 Add Supply Records:

To add a supply record, click the Add Record button and fill the information and click *Save* to add it or *Cancel* to return back.

Note: Supply records cannot be modified or deleted by any user. But can be added by either Admin or Supply and Design Manager.

4.6 Production Records:

To track the production records, information of factories and their production information like quantity and date. Select the Production Records option in the dashboard and the page will appear. It contains the list of all records with Factory Name, Model Name, Quantity, Production Date.

4.6.1 Add Production Records:

To add a production record, click the Add Record button and fill the information and click *Save* to add it or *Cancel* to return back.

Note: Production records cannot be modified or deleted by any user. But can be added by either Admin or Production Manager.

4.7 Factories:

To manage factories and their information, select the Factories option in the dashboard and the page will appear. It contains a list of all the factories with their location and production capacity.

4.7.1 Adding Factory:

Click the add factories option on the top right to add a new factory and fill the form with the details of the factories such as location and capacity and click *Save* to add it to the database or *Cancel* to exit.

4.7.2 Edit or Delete factory:

You can edit or remove the factory, by selecting the respective option present on the right side of every factory. Edit the factory by modifying details in the Edit Factory form. And click *Save* to make the changes or *Cancel* to return.

4.8 Employees:

To manage employees and their information, select the Employees option in the dashboard and the page will appear. It contains a list of all the employees with their name and role.

4.8.1 Adding employees:

Click the Add Employee option on the top right to add a new employee and fill the form with the details of the employee such as name and role , and click *Save* to add it to the database or *Cancel* to exit.

4.8.2 Edit or Delete employee:

You can edit or remove the employee, by selecting the respective option present on the right side of every employee. Edit the employee by modifying details in the Edit Employees form. And click *Save* to make the changes or *Cancel* to return.

4.9 Dealers:

To manage dealers and their information, select the Dealers option in the dashboard and the page will appear. It contains a list of all the dealers with their name,city and contact.

4.9.1 Adding dealers:

Click the Add Dealer option on the top right to add a new dealer and fill the form with the details of the dealers such as name,city and contact info , and click *Save* to add it to the database or *Cancel* to exit.

4.9.2 Edit or Delete suppliers:

You can edit or remove the dealers, by selecting the respective option present on the right side of every dealer. Edit the dealer info by modifying details in the Edit Dealers form. And click *Save* to make the changes or *Cancel* to return.

4.10 Sales:

To track the sales records, information of car sales from various dealers with data like dealer name, model name, quantity, sale date. Select the Sales option in the dashboard and the sales page will appear. It contains the list of all records with Dealer Name, Model Name, Quantity, Sales Date.

4.10.1 Add Sales Records:

To add a sales record, click the Add Record button and fill the information and click *Save* to add it or *Cancel* to return back.

Note: Sales records cannot be modified or deleted by any user. But can be added by either Admin or Sales Manager.

4.11 Manage users:

The admin user can add and manage users, assign them roles. Select the Manage Users option to open the page. Admin users can add, delete and edit users.

5. Production Dashboard:

Production Manager user can access the following limited functions and data:

- **Production records:** Track production records of the factories by car models.
- **Factories:** Manage factory locations and production capacity.
- **Employees:** Add, modify or delete employees records.

The use cases and functionalities remain the same as in the Admin dashboard.

6. Sales Dashboard:

Sales Manager user can access the following limited functions and data:

- **Dealers:** Manage dealers and their location information.
- **Sales:** Track car model sales transactions through dealers.

The use cases and functionalities remain the same as in the Admin dashboard.

7. Supply & Design Dashboard:

Supply and Design Manager user can access the following limited functions and data:

- **Car Models:** Add, modify or delete vehicle model specs, pricing and engine types.
- **Assign Parts:** Assign parts with required quantities to car models.
- **Parts:** Add, modify or delete parts inventory, pricing and their stock.
- **Suppliers:** Add, modify or delete suppliers and their contacts.
- **Supply records:** Track part supply transactions from the suppliers.